Analysis of Zoning Utilization Ratios in Newly Completed New York City Residential Buildings from 2012 to 2025.

Jeremy Gomez

Objective:

The goal of this project is to analyze the factors that lead developers of new residential buildings to maximize the allowable "Floor Area Ratio" of newly constructed buildings in New York that received certificates of occupancy between July 2012 and November 2025.

I will create the response variable "Maxed_Out" where the response is equal to "Yes" if the "Zoning Utilization Ratio" (built floor area ratio divided by the allowable floor area ratio) is greater than .9 and is equal to "No" if this ratio is less than .9.

$$Maxed_Out = "Yes"\ if, \frac{Built\ FAR}{Allowable\ FAR}\ \geq .9, Otherwise\ Maxed_Out\ =\ "No"$$

Methods:

I will use logistic regression model to predict whether a new residential building will have its zoning utilization ratio "Maxed Out" based on predictor variables borough, community board, lot size and zoning district.

To investigate temporal trends, I will apply non-parametric smoothing techniques (specifically Locally Estimated Scatterplot Smoothing, or LOESS) to the zoning utilization ratio. Non-parametric smoothing allows the model to adapt to local fluctuations in the data. This will enable me to visualize non-linear patterns, and comment on whether the intensity of land use in NYC has changed in response to major events in the NYC residential construction industry such as 2016 zoning quality text amendments, the COVID-19 pandemic, and the expiration of the 421-a Tax Exemption.

Data:

I will be working with 4 data sets all sourced from NYC Open Data, a public repository for NYC municipal data provided by the NYC Office of Technology and Innovation.

DOB Certificate of Occupancy

This data set is collection of all Certificates of Occupancy issued by the City of New York from July 2012 to March 2021; a Certificate of Occupancy states a building's legal use and/or type of permitted occupancy and is issued to new buildings once the NYC Department of Buildings inspects the completed building and deems it to be habitable.

DOB NOW: Certificate of Occupancy

Like the above dataset, this dataset is collection of all Certificates of Occupancy issued by the City of New York from March 2021 to November 2025, this data is in a separate database due to the migration of department of builds certificate of occupancy data from DOB Building Information System to DOB NOW in March of 2021. I will combine the certificate of occupancy data sets using R.

DOB JOB Application Filings

The DOB NOW Certificate of Occupancy dataset does not include a count of the proposed dwelling units in each building. Because of this I will need to join the DOB JOB Application Filings dataset to the DOB NOW: Certificate of Occupancy dataset to source this missing variable. The proposed dwelling unit variable is necessary for data cleaning purposes as I will filter out all nonresidential buildings by excluding all buildings that received a final certificate of occupancy with less than 1 residential unit, these buildings are usually non-residential structures such as garages and commercial buildings which will be excluded from the dataset.

Primary Land Use Tax Lot Output (PLUTO)

The Primary Land Use Tax Lot Output (PLUTO) data set is where we will source the Built FAR and Allowable FAR from. I will join this data to the combined certificate of occupancy dataset using the concatenated Borough, Block and Lot numbers, this serves as a unique identifier of each individual tax lot in New York City.